# SOHAM KAR

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## Technical Skills

**Programming, Data, and Frameworks:**

- *C, Python, Dart, Vue.js, Typescript, SQL, NoSQL, FastAPI, Pydantic, PostgreSQL, Redis, Flutter, FirebaseFirestore, Docker, Git, Linux Server Management, Nginx, Pandas, NumPy, SciPy, Matplotlib, Scikit-Learn, TensorFlow 2.0, PyTorch*

**Electronics & Prototyping:**

- *Arduino, ESP8266/ESP32, Sensors, IoT Systems, MQTT, 3D Printing, CNC, Laser Cutting, Soldering & Circuit Assembly*

**Design Software, and Engineering & Tools:**

- *AutoCAD, SolidWorks, Illustrator, Figma, STAAD.Pro, Android Studio, GRASS GIS, Blender*

## Professional Experience

**Rever Enterprise Platform — Inventory, Order & Business Management System** ***2025 – present***

*Independent Software Developer | Web Platform | Deployed | https://reverlife.in/home*

- Designed and developed a production-ready ERP platform to digitize inventory, order processing, distributor management, and financial workflows for a multi-level distribution business.
- Architected the complete backend using **FastAPI**, **PostgreSQL**, **Redis**, and **Docker**, implementing modular APIs(20+), database schemas and tables(15+), authentication(email and OTP), role-based access control, and business logic.
- Developed secure inventory, order, approval, and ledger management modules with hierarchical user permissions, audit trails, and timestamped transaction records.
- Built responsive administrative dashboards and user interfaces using **Vue.js**, translating complex business workflows into intuitive, scalable web applications.
- Optimized the system architecture for reliability and low operating costs by integrating **Firebase Authentication** for secure user management, **MSG91** for OTP and messaging services, and **Amazon S3** for object storage instead of hosting files on the application server.
- Migrated media and document storage from the VPS to **Amazon S3**, significantly reducing server disk usage and memory overhead, improving application stability and preventing resource exhaustion during concurrent file uploads. Containerized and deployed the platform on a Linux VPS using **Docker** and **Nginx**, implementing a lean infrastructure that minimized recurring costs while supporting production workloads.
- Collaborated directly with stakeholders to gather requirements, iterate on features, conduct client demonstrations, and support real-world testing prior to deployment.

**Co-Dafter — Co-working & Office Space Platform** ***2026 – present***

*Frontend Developer| Web Platform | in-Development | https://codaftar.com/*

- Developed a responsive business website using **Vue.js and Typescript** to showcase co-working spaces, managed offices, and leasing solutions.
- Built reusable **Vue.js components** and **responsive layouts** to provide a consistent user experience across desktop and mobile devices.
- Designed and integrated a **secure enquiry system with backend API connectivity**, enabling prospective clients to submit workspace enquiries.
- Collaborated with stakeholders to translate business requirements into clean, user-friendly interfaces and improve customer engagement.
- Structured the application with a modular component architecture to support future booking and workspace reservation features.

**Local Legends (formerly Match Up)** ***2024 – present***

*Co-founder | Mobile App & Website | MVP Deployed | https://locallegends.co.in/*

- Designed and developed a cross-platform football community platform to connect players with tournaments, trials, academies, venues, and local sporting events.
- Built the mobile application using **Flutter** and developed the backend with **FastAPI**, **Postgres, and Redis** implementing RESTful APIs, authentication, and scalable data models.
- Designed responsive user interfaces and user journeys, focusing on intuitive navigation and an engaging mobile-first experience.
- Integrated **Firebase Authentication**, cloud services, and backend APIs to enable secure user registration and account management.
- Planned and architected a scalable platform to support community growth, event management, and future social networking features.
- Led the complete product development lifecycle, including UI/UX design, system architecture, branding, deployment, and iterative feature development.

## Engineering Projects

**Built-Skin Innovations: Smart Facades** ***2024 – present***

*Idea-Stage Venture |Buildings and Deep Tech product |pitch deck link*

- Designed a modular, climate-responsive façade system in **SolidWorks** featuring scaffold-free installation and passive ventilation to improve building energy efficiency and reduce urban heat gain.
- Applied research on green façades and vertical building systems to develop the engineering concept, preliminary commercialization strategy, and unit economics.
- Selected for the **SRUJAN Pre-Seed Incubation Program** based on the project's innovation and commercial potential.
- Deferred prototype development after evaluating the substantial capital required for fabrication and large-scale validation, while continuing to refine the design and commercialization roadmap.

## Notable Publications

**Hydro Economy: Environmental Sustainability of Water and Wastewater Resources and Infrastructure**

- *April 2022 | **Published** | Book Title: Recent Developments in Sustainable Infrastructure (ICRDSI-2020)—GEO-TRA-ENV-WRM (pp.181-197) |https://doi.org/10.1007/978-981-16-7509-6_15*

**Assessing the Impact of Vegetation Coverage Ratio in a Floating Water Treatment bed of Pistia Stratiotes**

- *January 2021| **Published** | SN Applied Sciences | https://doi.org/10.1007/s42452-020-04139-2*

**Ecological Floating Bed for Decontamination of Polluted Water Bodies: Design, Mechanism & Performance**

- *September 2019| **Published** | Journal of Environmental Management | https://doi.org/10.1016/j.jenvman.2019.109550*

## Highlighted Achievement(s)

**Start-up SRUJAN Pre-Seed Incubation Program Selection** ***2024***

- Achieved recognition for start-up idea, Built-Skin Innovations, by getting selected for Start-Up SRUJAN seed support (S4) competitive pre-seed program by the Government of Gujrat at I-Hub Ahmedabad.

**INSDAG Civil Award Competition: 2019-2020** ***2021***

- Winners of **Institute For Steel Development & Growth**(INSDAG) Civil Award for the best innovative structural design of a Steel-Intensive Quarantine Center for COVID-19.

**Mitacs Globalink Research Internship (GRI)** ***2020***

- Recipient of **MITACS GLOBALINK SCHOLARSHIP** at the University of Concordia, Montreal, Canada.